Sop310

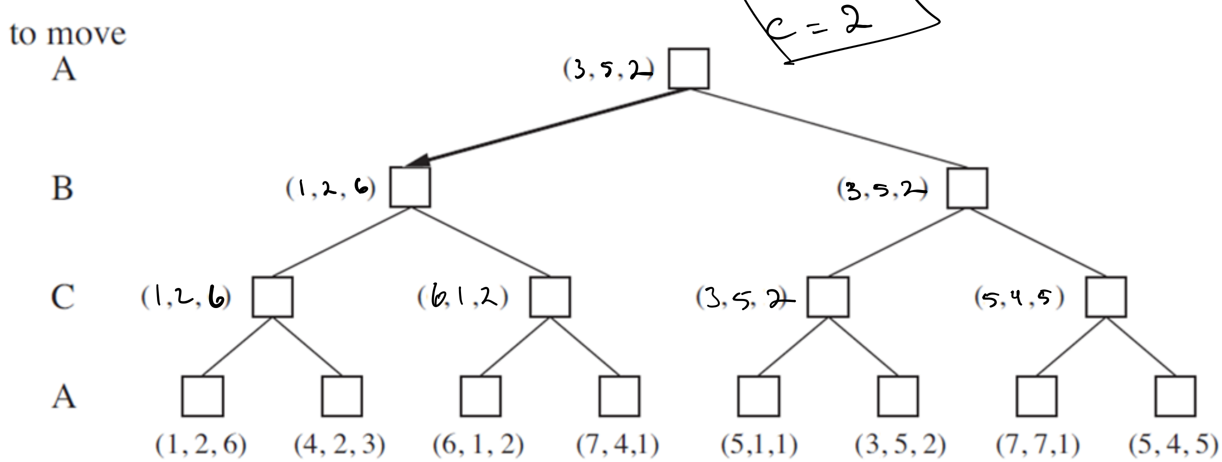
Question 1: Adversarial Search (6 points)

a) 5x5 Tic-Tac-Toe is a variation of the classic game played on a 5x5 grid, where the goal is to get five of your symbols in a row (horizontally, vertically, or diagonally) before your opponent does. How many nodes are there in the game tree for 5x5 Tic-Tac-Toe?

$\sum_{k=0}^{25} \frac{25!}{(25-k)} \approx \boxed{e \cdot 25!}$ nodes

b) Consider the following game tree for a 3-player game. Each leaf represents the score of the corresponding player (A, B, C) at the end of the game. If the Minimax function is applied, what will be the final score of each player?

A = 3
B = 5
C = 2



Question 2: Constraint Satisfaction Problems (6 points)

a) What is a binary constraint?

A Constraint that involves only two variables

b) Consider the constraint $Y = 2X^3$. Suppose the domains of X and Y are the sets of integers between 0 and 100 ($0 \leq X \leq 100$ and $0 \leq Y \leq 100$).

need some $y \leq 100$ satisfying $y = 2x^3$

1. What would be the reduced domain for X so that X is arc-consistent with respect to Y?

   $\cdot\ x = 3 \rightarrow 2(3)^3 = 2(27) = 54$ ✓

   $\cdot\ x = 4 \rightarrow 2(4)^3 = 2(64) = \cancel{128}$

   $X = \{0, 1, 2, 3\}$

2. What would be the reduced domain for Y so that Y is arc-consistent with respect to X?

   $Y = \{0, 2, 16, 54\}$

Question 3: Machine Learning (6 points)**:**

a) What types of learning are there in machine learning?

- Supervised
- unsupervised
- reinforcement

b) How does the output variable differ between regression and classification in machine learning?

outputs numerical values

outputs a specific class or category

Question 4: Gradient Descent (6 points)**:**

a) If there are 5 variables for each example in the data set, what is the equation for *h(X)* in multivariate linear regression? (Use the sigma symbol)?

$h(X) = w_0 + \sum_{i=1}^{5} w_i x_i$

b) Consider Least Mean Squared Error in linear regression for the following *(x,y)* training examples:

$h(1) = -2 + 5(1) = 3$; $h(2) = -2 + 5(2) = 8$

$h(3) = -2 + 5(3) = 13$; $h(4) = -2 + 5(4) = 18$

(1,2), (2,8), (3,15), (4,19)

If *h(x) = -2 + 5x*, calculate *J(w)*.

$(3-2) = 1$; $(8-8) = 0$; $(13-15) = -2$; $(18-19) = -1$

Note: *J(w)* $= \frac{1}{2n}\sum_{i=1}^{n}\left(h_w(x^i) - y^i\right)^2$

$1^2 = 1$; $0^2 = 0$; $-2^2 = 4$; $-1^2 = 1$

$1+0+4+1 = 6$; $J(w) = \frac{1}{2(4)}(6) = \frac{6}{8} \boxed{= .75}$

Question 5: Decision Tree (6 points)**:**

a) Fill in the blanks.

In a Decision Tree representation:

Each internal node tests an attribute

Each branch corresponds to a potential value of the attribute

Each leaf node assigns a classification

b) In a binary classification, if one quarter of the data belongs to one class and the rest belongs to another class, calculate the entropy of the data to three decimal places.

$p_1 = 1/4$

$p_2 = 3/4$

Note: *log2(1/4) = -2* and *log2(3/4) = -0.415*

entropy $= -\frac{1}{4}\log_2\left(\frac{1}{4}\right) - \frac{3}{4}\log_2\left(\frac{3}{4}\right) = -\frac{1}{4}(-2) - \frac{3}{4}(-.415)$

$= .5 + .311 = \boxed{.811}$